

# Oleksii Odarchuk

## Backend Developer (Go) · Creator of the Piton language

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Performance-obsessed engineer with a product mindset. 1st place at BEST Hackathon 2026, 3rd at Mate Hackathon, invited to DOU Day 2026 as a Student with High Potential. Builds real products fast — from hackathon MVPs to production Go services.

### 🚀 WINS & HACKATHONS

#### 🏆 1st Place — Lead Backend & Sensor Fusion

April 18–19, 2026

*BEST Hackathon 2026 — GPS-less Drone Trajectory*

- Algorithm to **reconstruct a UAV's trajectory without GPS** — from IMU, barometer and other inertial-sensor logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C inline for the hot numerical kernels.
- 1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

#### 🏆 3rd Place — Backend Engineer

March 14, 2026

*Mate Hackathon (50 participants, 10 teams)*

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for marketing trigger analysis + Nano Banana 2 (fal.ai) for creative generation; backend routed all AI calls to cloud APIs, sustaining throughput under hackathon load.
- Beat teams with 5+ years of experience by focusing on a stable MVP — **3rd of 10 teams**.

#### Security Researcher → Job Offer

2026

*College Blockchain System "Student Hryvnia"*

- Discovered and exploited a critical vulnerability: frontend capped transaction amounts, **backend had no such check** — enabling arbitrary negative-value transfers. Reported responsibly; received a job offer from administration.
- Reverse-engineered the official JS miner through Webpack bundles — then built a faster Go replacement with a goroutine worker pool (SHMiner — p. 2).

#### 📄 DOU Day 2026 — Student with High Potential

Personally invited by the organisers with a free ticket — a dedicated track for promising student developers.

### 🔧 TECH STACK

Languages: **Go** **Python** **C**

**TypeScript**

Backend: **Gin** **Fiber** **REST**

**WebSockets**

**Goroutines**

**JWT**

**OAuth 2.0**

Databases: **PostgreSQL** **Redis**

**ClickHouse**

Frontend: **Svelte** **Wails**

**Tailwind**

AI / APIs: **GPT-4o** **Fal.AI**

DevOps: **Docker** **Linux (Arch)**

**Git**

**TinyGo**

**Bruno**

**Typst**

### 🎓 EDUCATION & LANGUAGES

#### Network Technologies — IT

College NULP, 2nd year

2024 — present

#### IT Class — Science Lyceum, Rivne

2022 — 2024

**Ukrainian** — native · **English** — B1

 **Piton — My own programming language**[github](#) → [playground](#) → [docs](#) →*Ukrainian transliterated language + interpreter + flowchart generator*

Interpreted language with an authentic Ukrainian syntax (in transliteration). Written from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, Turing-complete. Flowcharts are drawn by a built-in **D2**-based engine. Web playground: interpreter compiled to **WebAssembly via TinyGo** ( 220 KB vs. 24 MB with stock Go).

[Go](#) [TinyGo](#) [WebAssembly](#) [AST](#) [D2](#) [Lexer + Parser](#) [Nix Flakes](#)**OPEN SOURCE SDK****go-monobank-sdk —** [github](#) →  
**Go SDK for every monobank API**

Full coverage of **personal**, **provider** (with monoKEП), **business** for legal entities, **acquiring** (subscriptions, monopay button, split-receivers, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in `http.Handler`, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

[Go](#) [ECDSA](#)[HMAC-SHA256](#)[OpenTelemetry](#) [iter.Seq2](#)**SDK CONSUMER****monokasa — Telegram** [github](#) →  
**Ticket-Selling Bot**

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

[Go](#) [SQLite](#) [telebot](#)[PDF + QR](#)[go-monobank-sdk](#)**1000× FASTER****SHMiner — Desktop** [github](#) →  
**Crypto Miner**

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

[Go](#) [Svelte](#) [TypeScript](#)[Wails](#) [WebSockets](#)[Goroutines](#)**150+ USERS****AbitAssistant Bot** [github](#) →

Telegram bot helping Ukrainian students track university admission rankings. Reverse-engineered closed third-party APIs to fetch live competition data. **150+ active users in first year**, marketing campaign planned for 2026.

[Python](#) [Aiogram](#)[PostgreSQL](#) [BeautifulSoup](#)[Pandas](#) [Docker](#)**LinkShortener Bot** [github](#) →

Telegram URL shortener with QR generation and geo analytics. PostgreSQL for storage, Redis for redirect caching, ClickHouse for real-time click analytics (country, city, platform). Docker, GeolP2.

[Go](#) [PostgreSQL](#) [Redis](#)[ClickHouse](#) [Docker](#)[Telegram API](#)**SAVES HOURS****rombik — Flowchart** [website](#) →  
**Generator (SaaS)**

A SaaS I designed and run: generates state-standard (DSTU) flowcharts from **Python, C, C++, C#, Java and Pascal** code, saving hours of manual drawing for academic papers. **Go engine**, web editor (Svelte 5), a public **HTTP API**, a **CLI**, an **MCP server** and an AI skill; export to SVG/PNG/PDF/Typst/Excalidraw. Free PNG, no sign-up.

[Go · Gin](#) [tree-sitter](#)[API · CLI](#) [MCP](#)